



Prepared with Synoptic

Evidence and Argument Tables

Grounding on Boolarong Bank, Merrandi Channel, Kanooka Heads

The evidence, tests and reasoning behind the investigation's findings.

Date of issue

09/10/2026

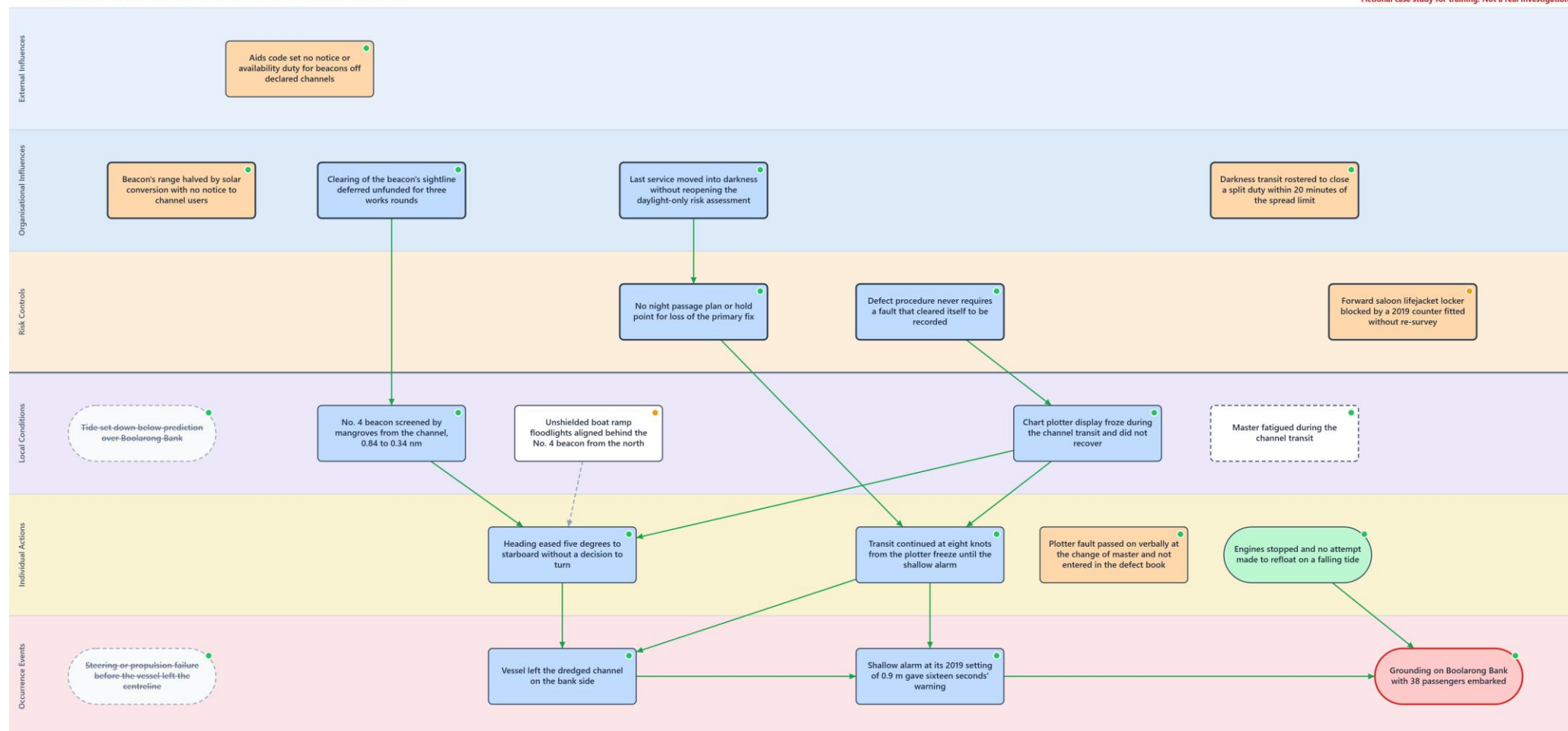
Contents

AcciMap	3
How to read these tables	4
External Influences	5
Aids code set no notice or availability duty for beacons off declared channels	5
Organisational Influences	6
Beacon's range halved by solar conversion with no notice to channel users	6
Clearing of the beacon's sightline deferred unfunded for three works rounds	7
Last service moved into darkness without reopening the daylight-only risk assessment	8
Darkness transit rostered to close a split duty within 20 minutes of the spread limit	9
Risk Controls	10
No night passage plan or hold point for loss of the primary fix	10
Defect procedure never requires a fault that cleared itself to be recorded	11
Forward saloon lifejacket locker blocked by a 2019 counter fitted without re-survey	12
Local Conditions	13
Tide set down below prediction over Boolarong Bank	13
No. 4 beacon screened by mangroves from the channel, 0.84 to 0.34 nm	14
Unshielded boat ramp floodlights aligned behind the No. 4 beacon from the north	15
Chart plotter display froze during the channel transit and did not recover	16
Master fatigued during the channel transit	17
Individual Actions	18
Heading eased five degrees to starboard without a decision to turn	18
Transit continued at eight knots from the plotter freeze until the shallow alarm	19
Plotter fault passed on verbally at the change of master and not entered in the defect book	20
Engines stopped and no attempt made to refloat on a falling tide	21
Occurrence Events	22
Steering or propulsion failure before the vessel left the centreline	22
Vessel left the dredged channel on the bank side	23
Shallow alarm at its 2019 setting of 0.9 m gave sixteen seconds' warning	24
Grounding on Boolarong Bank with 38 passengers embarked	25

Fictional case study for training. Not a real investigation.

AcciMap

Grounding on Boolarong Bank, Merrandi Channel, Kanooka Heads

Rendered 08/10/2026, 2:30:36 pm
Fictional case study for training. Not a real investigation.

Fictional case study for training. Not a real investigation.

How to read these tables

Each table is one object from the analysis map. An object is a factor: an event (something that happened) or a condition (a state of affairs that existed at the time). Each factor is worded as a proposition, a statement that is either true or not, which the three tests then examine.

The tests run in order. Existence asks whether the factor existed or occurred. Influence asks whether it influenced this occurrence. Importance asks whether it is worth addressing regardless. A later test only means something once the earlier one is settled: Influence is tested once Existence has passed, and Importance where Influence failed. A test not yet reached reads Not tested.

The shaded row opens each test with its result (Pass, Fail, Inconclusive or Not tested) and the confidence behind it. Confidence follows the probability scale: virtually certain, extremely likely, very likely, likely, more likely than not, about as likely as not, unlikely, very unlikely, extremely unlikely and exceptionally unlikely, with the percentage in brackets. High confidence passes the test, low confidence fails it, and the middle is Inconclusive.

Under the result, Argument For sets out the case that the factor, as worded, is true for that test, and Argument Against the case that it is not. Both are always read against the factor as worded, never against the result, so a failed test still shows its case for. Reasoning says which side carried and why.

Nil means nothing was recorded on that side. A test argued both ways shows a case on each side, and the reasoning says which one carried. A test with nothing recorded shows only its result row: no argument was written for it, so its result is not supported by this record.

External Influences

Aids code set no notice or availability duty for beacons off declared channels

Existence:	Pass, virtually certain (100%)
Argument For: WAN-2 section 4.2 applies an availability target of 99.0 per cent, an obligation to record availability, and an obligation to notify a change of character, position or range by notice to mariners, only where an aid stands on a declared shipping channel. The Merrandi Channel is not one, so section 4.3 applies instead and requires only that the responsible authority can demonstrate an inspection interval it has set itself. Neither 4.3 nor the services agreement that sets that interval asks for availability to be recorded or for any user of the channel to be told of a defect, a deferral or a change. Section 4.1, which requires every aid to be maintained so that it performs the function for which it was established, has no channel threshold and is not part of this proposition. Argument Against: Nil Reasoning: The code's text and the channel's status are documentary and not in dispute.	
Influence:	Fail, unlikely (30%)
Argument For: Had 4.2 applied, the range change of 11/2025 would have gone out as a notice to mariners before it took effect, and the Authority would have been recording the beacon's availability. Those are the only two mechanisms in the code that would have carried anything about the No. 4 beacon to the people using the channel, and neither existed for it. Argument Against: Neither mechanism reaches this occurrence. A notice of the range change would have told the operator about a light that was screened over the stretch where the vessel left the channel, and a better known light is no more visible through a mangrove canopy; the solar conversion object fails on the same ground. An availability figure measures whether a light is lit, and this one was lit the whole time. The screening itself was not left because of the code: 4.1 applies to every aid whatever the channel's status, the Authority's own register has recorded the screening since 09/2023, and what kept WO-118 unfunded was the services agreement's 50 m vegetation scope, which put the work in the general waterways vote. Reasoning: Argued and failed at 30%. The evidence does not leave the question open: each mechanism the code withholds runs into a fact the record settles against it. What remains is that nobody using the channel was ever told anything about its aids, which is an Importance question.	
Importance:	Pass, likely (80%)
Argument For: It did not influence this occurrence and it is still worth addressing. On a channel carrying a 96-passenger ferry in darkness, the code gives the operator no right to be told when a beacon's range, character or position changes and no figure for how often the aids work. The solar conversion reached nobody for eight months for exactly that reason, and the next change, to a light that is not screened, would reach nobody either. Argument Against: Nil Reasoning: A standing gap in what the users of an undeclared channel are told, which the Authority can close by its own practice whatever the code requires.	

Organisational Influences

Beacon's range halved by solar conversion with no notice to channel users

Existence:	Pass, virtually certain (100%)
Argument For: The No. 4 beacon was converted from a mains supply to a solar lantern in 11/2025 under the Council's waterways energy programme. The replacement lantern is rated at a nominal 3 nautical miles against the 6 of the lantern it replaced; the character, Fl G 2.5s, did not change. A notification was lodged with the hydrographic service on 04/02/2026 and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator using the channel was informed. The investigator's measurement from the port workboat on 18/07/2026 gives a luminous range of 2.6 nm. Tarrakin Ferries' passage plan appendix, issue 2 and current, records the No. 4 beacon at 6 M. Argument Against: Nil Reasoning: The conversion, the lantern rating, the hydrographic lodgement and the absence of any local notice are all documentary, and the range was measured independently.	
Influence:	Fail, unlikely (20%)
Argument For: The master called the beacon 'a nothing' in front of the town and the new floodlights, and said 'It used to be a proper light. It was a mains light.' For at least the last 37 seconds of his search the light was clear of the islet and not yet seen, and a weaker light against a bright background is harder to pick up in that time. The passage plan he was working to still gave the No. 4 beacon at 6 M. Argument Against: Over the half mile in which the vessel left the channel the beacon was screened by the canopy, and a screened light is not made visible by being brighter. The master already knew the light was weaker, in his own words, so a notice would have told him nothing he did not know. Range is not the limit at these distances either: the search ran from 0.47 nm to 0.25 nm, well inside even the measured 2.6 nm. Nothing in the record shows the plan's 6 M figure was used for anything on the night. Reasoning: Argued and failed at 20%. The one mechanism left is the light's conspicuity against the floodlights in the seconds after it cleared the islet, which is the floodlight object's question and stays unresolved there; this object adds nothing to it that the record can carry.	
Importance:	Pass, likely (85%)
Argument For: It did not influence this occurrence and it is still a finding. A light on a 96-passenger route had its nominal range halved, the chart correction was still pending eight months later, and no user of the channel was told. The operator's passage plan described a light that was no longer there. On a night when the light is not screened, or with another aid changed the same way, a master working to the plan is looking for something twice as strong as what is there. Argument Against: Nil Reasoning: A change to an aid that reached no user of the channel; answered in part by LNM-2026-14 and the release of the register (sa-3).	

Clearing of the beacon's sightline deferred unfunded for three works rounds

Existence:	Pass, virtually certain (100%)
Argument For: Work order WO-118, raised 09/2023, records mangrove regrowth on Merrandi Islet screening the No. 4 beacon from the northern approach between about half a mile and a quarter mile, recommends trimming to 2 m on a three-yearly cycle, and estimates \$18,500. It was deferred at the 2023, 2024 and 2025 works rounds, each entry reading 'defer, unfunded, refer next round'. WO-176, raised 03/2026, records the same sightline as 'now fully closed on the northern approach'. The Authority's file note against WO-176 records that the islet is 240 m from the beacon and so falls outside the 50 m scope of schedule 2.4, and that the work therefore competes for the general waterways vote rather than the aids to navigation vote. The 2019 aerial survey gives the canopy at 2.1 m; it measured 5.8 m on 18/07/2026 against a lantern height of 4.2 m. Argument Against: Nil Reasoning: Documented in the Authority's own register and measured on site.	
Influence:	Pass, very likely (90%)
Argument For: WO-118 recommended trimming to 2 m and a three-yearly cycle. The canopy grew from 2.1 m in 2019 to 5.8 m in 07/2026, about half a metre a year, so a trim funded at the 2023 round would have left it near 3.5 m on the night, under the 4.2 m lantern, and the beacon open to the channel over the stretch where it was screened. Argument Against: Nil Reasoning: The step from the deferral to the screening is arithmetic on the Authority's own figures: funded in 2023, the canopy stands about 0.7 m below the lantern on 07/07/2026 rather than 1.6 m above it. The 10% allows that the growth may not have been even over the seven years, since the register gives two measurements and the work order's description and nothing in between.	
Importance:	Not tested

Last service moved into darkness without reopening the daylight-only risk assessment

Existence:	Pass, virtually certain (100%)
Argument For: RRA-07 issue 2 (03/2023) is the current route risk assessment for the Merrandi Channel. It carries the operating restriction 'daylight transits only', and states that a night transit requires the General Manager Operations' approval and a specific passage plan. CHG-2026-014, approved by the Operations Manager on 22/04/2026, moved service TF18 from 16:15 to 17:35 for the winter timetable from 01/06/2026, assessed the change as 'timetable change only', recorded 'no new hazard identified' and answered 'No' to route risk assessment review required. RRA-07 was not reopened and no issue 3 exists. No night passage plan was written and the General Manager Operations' approval was neither sought nor given. TF18 does not run on Sundays or public holidays, and ran 31 times in darkness between 01/06/2026 and the occurrence, which was the thirty-first. Argument Against: Nil Reasoning: Both documents are current controlled copies and neither is in dispute: RRA-07 issue 2 carries the daylight restriction and names the approver, and CHG-2026-014 carries the approval, the 'timetable change only' assessment and the 'No' against route risk assessment review. There is no issue 3 and no night passage plan. What the Operations Manager understood the restriction to mean goes to why the change was approved that way, not to whether it was, and that uncertainty is carried under Influence.	
Influence:	Pass, likely (85%)
Argument For: RRA-07 named both the approver and the plan a night transit needed. A review of it against a night operation would have reached the unlit reach between the No. 2 and No. 4 beacons, which the assessment already names as a hazard, and could not have closed without a night passage plan. Instead the change put a 96-passenger vessel through a 40 m channel in darkness 31 times, through a process that asked whether the change was new rather than whether the assessment still held. Argument Against: A reopened assessment might have produced a night plan that said no more than the daylight one does, with no hold point in it, and the vessel would have met the frozen plotter in the same way. The Operations Manager's account of what he understood the restriction to require of him has been given verbally and has not been taken as a formal interview, so why the change was assessed as a timetable change only is not yet established. Reasoning: Argued. Accepted at 85%: the restriction names the plan, so a review that read it could not have ended without one, and a night plan written for this channel is the document most likely to say what to do when the primary fix is lost. The Against goes to how good that plan would have been, not to whether there would have been one.	
Importance:	Not tested

Darkness transit rostered to close a split duty within 20 minutes of the spread limit

Existence:	Pass, virtually certain (100%)
Argument For: In the winter timetable TF18 closes the afternoon half of a split duty, 05:40 to 08:40 and 13:00 to 19:20, worked by the master on that line on every day of a 12 days on, 2 days off tour. That is 9 hours 20 minutes of duty across a 13 hour 40 minute spread, against the operator's limits of 10 hours of duty and 14 hours of spread, so the darkness transit falls in the last hour and a quarter of a spread that ends 20 minutes inside the limit, on days nine to twelve of the tour as on the rest. No relief master is rostered; a master who reports unfit is covered by extending another master's tour. On 07/07/2026 the master was on day nine, and had made 26 of the 30 darkness transits before it. Argument Against: Nil Reasoning: The roster, the duty record and the operator's limits are documentary.	
Influence:	Fail, unlikely (25%)
Argument For: The roster put the master through a darkness channel transit at the end of a split duty on day nine of twelve, after 6 hours 15 minutes of sleep, and he described himself as tired. The decisions the occurrence turned on, not stopping when the plotter froze and easing back rather than stopping once he saw the beacon dead ahead, are the kind fatigue is expected to degrade. Argument Against: A roster can reach this occurrence only through the master's fatigue, and the recorded evidence gives no sign of it on the night (see the fatigue object). Helm work held the centreline to within 3 m for the first two minutes in the channel; the response to the shallow alarm was 4 seconds; the radio work after the grounding was complete and correct at the first attempt; and the decision not to refloat was made in about thirty seconds and was right. Not stopping when the plotter froze is what both masters said nothing in any document asks of them, and the outgoing master, off a morning shift, gave the same account of the handover and the beacon. Reasoning: Argued and failed at 25%. Failed on the evidence, not dismissed: the roster's only path to this occurrence runs through a fatigue the record does not establish.	
Importance:	Pass, likely (85%)
Argument For: It failed Influence for this occurrence and remains a finding worth acting on. The winter timetable puts the last darkness transit of the day at the end of a split duty whose spread runs to within twenty minutes of the operator's own limit, on every day of a twelve-day tour, with a fatigue diary the operator asks for and never reads, and no relief master to break a tour when somebody reports unfit. Nothing about that changes because it did not bite here. Argument Against: Nil Reasoning: A standing exposure built into the winter roster rather than a one-day event; it recurs on every darkness transit of the timetable.	

Risk Controls

No night passage plan or hold point for loss of the primary fix

Existence:	Pass, virtually certain (100%)
Argument For: OPS-3 3.4 requires the channel transit to be made on the charted centreline, monitored on the chart plotter, with the No. 2 and No. 4 beacons identified before passing No. 1. Appendix A, issue 2 and current, names the plotter as the primary means of position fixing and, as the secondary, the visual transit of the No. 4 beacon with the town wharf leading lights, and closes with the note that the plan is written for the daylight transit and that night transits are not scheduled. There is no night passage plan for the Merrandi Channel, no guidance anywhere on what to do if the plotter is lost during the transit, and no abort or hold point named in the plan at all. The leading lights half of that secondary means bear 195 degrees but lie about 1.9 nm beyond the No. 6 beacon and do not resolve as two lights until the vessel is inside the channel's southern third, so at the point in the transit where the plotter was lost the beacon was the whole of it. OPS-3 3.6 requires a reduction of speed and consideration of anchoring, but only in reduced visibility, which this was not. Argument Against: Nil Reasoning: The manual text and the appendix are documentary, and the absence of a night plan was confirmed by the operator.	
Influence:	Pass, very likely (90%)
Argument For: This is the operator's own defence against exactly this sequence, and it was never written. A hold point costs nothing on a flat night with no traffic, and every master asked said they would have used one. Argument Against: Nil Reasoning: The master had 114 seconds between the freeze and crossing the channel edge, and a crash stop begun before about 18:11:40 would still have stopped the vessel short of the contact point, on flat water with no other traffic. A named hold point would have applied throughout. The 10% allows that a master who did not think of stopping unprompted might not have applied a written rule under the same pressure.	
Importance:	Not tested

Defect procedure never requires a fault that cleared itself to be recorded

Existence:	Pass, virtually certain (100%)
Argument For: OPS-9 9.1 requires immediate reporting only of a defect rendering the vessel unseaworthy or affecting a statutory item. The chart plotter is not a statutory item on this vessel on this route, although it is the primary means of position fixing named in the passage plan. 9.2 puts all other defects in the vessel defect book at the end of the shift in which they are found, which for a fault that has come and gone by then is no trigger at all, and the manual nowhere says whether such a fault is a defect. 9.3 requires the outgoing master to brief the relieving master on any defect, and requires nothing about that brief to be written down and nothing of the relieving master afterwards. Both masters described the same reading independently: 'The book is for things that are broken when you hand over. If it's broken you write it, if it came good you say it.' The book's eleven entries in the 90 days before the occurrence are all faults present when they were written. Argument Against: Nil Reasoning: The manual text is documentary: no provision in OPS-9 requires a fault that has cleared itself to be recorded, and both masters read it that way without being asked.	
Influence:	Pass, likely (75%)
Argument For: On the outgoing master's account the plotter froze three or four times on this vessel in the year before the occurrence, it froze on the relieving master a month earlier, and it froze twice on the morning of 07/07/2026, and none of those freezes was ever written down because each had cleared itself. Recorded, they would have reached the Operations Manager's weekly review and the maintenance system, and from there the contractor, who identified cause code 0x51 against the 11/2024 firmware in one bench test. The fix for the fault that froze the display at 18:07 was never applied because no freeze was ever on paper. Argument Against: The chain from a written entry to an applied fix runs through somebody ashore finding the manufacturer's bulletin, and the operator held no subscription to the bulletins, so a recorded freeze might have been reviewed at the end of the week, noted and left. Nothing establishes how many recorded freezes it would have taken for the unit to be sent to the contractor, or how quickly. Reasoning: Argued. Accepted at 75%: a year of recorded freezes on the vessel's primary means of position fixing is unlikely to have gone to nobody, and a single bench test was enough to name the fix. The Against goes to how quickly, which the record cannot answer, and the report should carry both.	
Importance:	Not tested

Forward saloon lifejacket locker blocked by a 2019 counter fitted without re-survey

Existence:	Pass, virtually certain (100%)
Argument For: Raised by the deckhand, who had to ask three passengers to move away from the kiosk counter and pass behind it through a 0.6 m gap to reach the forward saloon locker while issuing lifejackets. The survey on 16/07/2026 found the locker face cannot be reached from 22 of the 34 forward saloon seats without passing behind the counter, and that with people standing at the counter the inboard gap is the only route. The muster placard directs passengers to a locker they cannot see from their seats. The arrangement approved at build in 2011 had an open bench in that position with the locker face clear; the counter was fitted in a 2019 interior refit carried out under a minor modification notice that did not trigger a re-survey of the safety equipment arrangement. The operator says the same counter is fitted to its two other 24 m vessels. Argument Against: Nil Reasoning: Measured on the vessel, with the refit notice and the 2011 approved arrangement on file.	
Influence:	Fail, unlikely (10%)
Argument For: Forty lifejackets behind a counter, in a saloon of 34 seats, with a placard pointing at something nobody can see, is a defence that is not there when it is needed. Argument Against: Nil Reasoning: The vessel grounded with the locker where it was and would have grounded with it anywhere. All 38 passengers were in lifejackets within about eight minutes and ashore within 74. There is no path by which this affected the grounding or its outcome.	
Importance:	Pass, very likely (90%)
Argument For: It did not influence this occurrence and it is still a finding. The vessel was aground, upright, stable and not sinking; the muster was unhurried and took a couple of minutes longer than it should have. Change one thing (a fire, a capsized risk, a night evacuation into the water, or a deckhand injured in the impact) and a locker the passengers cannot reach without crew help becomes the difference between a saloon that equips itself and one that waits. The locker is still behind the counter on this vessel, and on the operator's answer of 16/07 it is behind the same counter on the other two. Argument Against: Nil Reasoning: A degraded escape defence still in place, created by a refit that triggered no re-survey; whether it is fleet-wide waits on the operator's and the regulator's answers.	

Local Conditions

Tide set down below prediction over Boolarong Bank

Existence:	Fail, very unlikely (5%)
Argument For: Hypothesis tested because the vessel grounded three hours into a falling tide with a two-day westerly behind it, and a set-down of a few tenths would have mattered. It fails, and what refutes it is measurement rather than argument. The Port Authority's gauge at Kanooka Heads read 0.91 m at 18:12. The height predicted for that minute was 0.94 m, so the stand was 0.03 m under the tide table, and the gauge's largest departure from the table in the twelve hours to 20:00 was 0.07 m. With the bed at the contact point surveyed at 0.50 m below chart datum there was 1.41 m of water over it, against a static draught of 1.35 m and a calculated squat of 0.18 m at 6.8 kt. The vessel touched because it was 54 m outside a channel dredged to 3.2 m, not because the water was lower than the chart and the tide table said it would be. Argument Against: Nil Reasoning: The gauge record and the tide table are independent of each other and of the vessel, and they agree to within 0.03 m at the moment of the grounding, which is the ordinary scatter between a gauge and a table rather than a set-down. Taken at face value it changes nothing: at the predicted height there would have been 1.44 m of water against an effective draught of 1.53 m, and the vessel would have touched at the predicted tide as well. The 5% is the standing of the hypothesis and not of those readings, which are themselves settled.	
Influence:	Not tested
Importance:	Not tested

No. 4 beacon screened by mangroves from the channel, 0.84 to 0.34 nm

Existence:	Pass, virtually certain (100%)
Argument For: The range and visibility check on 18/07/2026, made from the port workboat on the inbound track and at the same time of night, found the No. 4 beacon is not visible at all from the channel between 0.84 nm and 0.34 nm north of it. The sightline over that stretch passes through the mangrove canopy on Merrandi Islet, which stands about 240 m up-channel of the beacon in the shoal strip between the dredged western edge and the bank; the canopy measured 5.8 m against a lantern height of 4.2 m. Beyond 0.84 nm the sightline passes west of the islet and inside 0.34 nm it passes east of it. The register has recorded the same screening since 09/2023 and recorded the sightline as 'now fully closed on the northern approach' in 03/2026. Both masters interviewed raised the beacon unprompted, and neither has ever reported it. Argument Against: Nil Reasoning: Measured on site from the inbound track at the same time of night, and recorded in the Authority's own register for nearly three years. Eleven days is no reason to doubt a mangrove canopy. The doubt the measurement does carry is geometric, because the vessel was west of the inbound track, and that goes to Influence.	
Influence:	Pass, likely (85%)
Argument For: With the plotter frozen the beacon was the only means of fixing left, and it was screened while the new heading was held. The bridge audio has 'Right. Where's four' at 18:08:31 and 'There's four' at 18:10:02, 91 seconds apart; the vessel was 0.47 nm north of the beacon when he began looking, crossed the channel's western edge at about 0.40 nm, inside the screened arc, and was 0.25 nm north when he called it. Seen on the starboard bow where it belongs, the light would have shown the vessel drawing west within a minute of the turn, while it was still in the channel. Argument Against: The heading change began at 18:07:06, while the master was at the frozen plotter and before he started looking for the beacon, so the beacon's absence cannot have caused the turn, only its going unnoticed. The light was also clear of the islet for at least the last 37 seconds of the search before he called it (0.34 nm at about 18:09:25, called at 0.25 nm), and for longer from the vessel's own track west of the inbound one, so for part of the search there was something to see and it was not seen. Reasoning: Argued. Accepted at 85%: the turn was made at the plotter, but it was held and went unnoticed because the one reference that would have shown it was screened over the minute and a half that mattered. The 37 seconds are the floodlight object's question and do not undo the screening before them.	
Importance:	Not tested

Unshielded boat ramp floodlights aligned behind the No. 4 beacon from the north

Existence:	Pass, virtually certain (100%)
Argument For: The survey on 18/07/2026 found 14 unshielded LED floodlights on 12 m columns at the new boat ramp, each presenting about 4,800 candela toward the channel, commissioned on 22/05/2026, ten days before the winter timetable began. From 0.5 nm north of the beacon the array lies behind it within 4 degrees of the same bearing and within 1.5 degrees of the same elevation, and it is continuously lit while the beacon flashes green for 0.4 seconds every 2.5 seconds. The Port Authority's aids to navigation officer was not consulted on the development and no assessment against the channel's aids was required or made. Argument Against: Nil Reasoning: Measured from the port workboat on the inbound track. The floodlights, their output, their shielding and their alignment are not in dispute.	
Influence:	Inconclusive, about as likely as not (45%)
Argument For: The master described looking 'into a wall of light' and called the beacon 'a nothing' against the town and the new car park, and the outgoing master independently called it 'washed out altogether from up the top'. For at least the last 37 seconds before the master called the beacon it was clear of the islet and not seen, which is the stretch in which the floodlights, rather than the canopy, stood between him and the light. Argument Against: Over most of the search the beacon was screened by the canopy and would not have been visible against a black background either. The heading change began at the plotter, before the search, so the floodlights cannot have caused the turn. No measurement exists of what could be seen on the night, only a survey eleven days later, nothing establishes a detection threshold for a 0.4 second green flash against that background, and the master himself said twice that he could not tell whether the lights were why he could not see the beacon. Reasoning: Inconclusive, and recorded as inconclusive rather than dropped. The alignment is real and the masters' accounts are consistent and independent; whether the floodlights delayed the sighting in the seconds after the light cleared the islet, and so the moment the held heading was noticed, cannot be established either way on the evidence available. The report owes the reader this reasoning rather than silence, and the Authority owes the channel a lighting assessment however this test lands.	
Importance:	Not tested

Chart plotter display froze during the channel transit and did not recover

Existence:	Pass, virtually certain (100%)
Argument For: The plotter's internal event log records the display process not responding at 18:07:06 local with no accompanying reset, and the same cause code 0x51 at 10:42 and 11:17 that morning, where the unit recovered itself. The bridge audio matches to the second: 'Oh, come on' at 18:07:11, repeated screen taps at 18:07:26, and '(unintelligible) frozen again' at 18:07:44. The master's account is that the chart picture remained on the screen with the vessel symbol stationary and the display not accepting touch input. The unit was bench tested for 46 hours on 21/07/2026 with the same antenna and cabling; the fault did not recur and no defect was found. The contractor identified cause code 0x51 as known to the manufacturer and addressed by firmware issued in 11/2024, which this unit had never had applied. Argument Against: Nil Reasoning: Three independent records place the freeze at 18:07:06 and none of them is contradicted: the unit's own event log, the bridge audio to the second, and the master's account. That the 46 hour bench test could not reproduce the fault and found no physical defect goes to what caused the freeze, which is a question for the firmware and the maintenance chain, not to whether the display froze during the transit. The doubt was carried here until 04/10/2026 and has been moved to where it belongs.	
Influence:	Pass, extremely likely (95%)
Argument For: This is the moment the transit stopped being the one in the passage plan. Everything the master did afterwards he did without the primary means of position fixing the plan told him to use. Argument Against: Nil Reasoning: The vessel held the centreline while the plotter worked. The heading change began at the second of the freeze, while the master was at the screen, and his search for the beacon began 85 seconds after it, with no position fix of any kind in between.	
Importance:	Not tested

Master fatigued during the channel transit

Existence:	Inconclusive, about as likely as not (50%)
Argument For: Day nine of a twelve-day tour, on a split duty of 05:40 to 08:40 and 13:00 to 19:20 with a 13 hour 40 minute spread, the channel transit falling in its last hour and a quarter. The master's own fatigue diary gives 6 hours 15 minutes of sleep the night before, and in interview he said 'I was tired. I'd be lying if I said otherwise.' Having seen the beacon dead ahead at 18:10:02 and known he was off the line, he eased the vessel back to port rather than stopping, on his account because he took himself to be just outside the edge when he was nearly 40 m west of the centreline, which is the kind of misjudgement fatigue is expected to produce. Argument Against: In the same answer he said 'I wasn't fighting to stay awake and I wasn't slow', and the recorded performance agrees with him. Helm work held the centreline to within 3 m for the first two minutes in the channel; the response to the shallow alarm was 4 seconds; the radio work after the grounding was complete and correct at the first attempt; and the decision not to refloat was made in about thirty seconds and was right. Six and a quarter hours is what he says is normal for him on a split, and nothing else in the record bears on his state. Reasoning: Inconclusive at 50%, and recorded as tested rather than dropped. The duty and the sleep make fatigue plausible and his own account admits tiredness; the recorded performance shows no sign of it. No detailed fatigue assessment was made, because whatever it found the object would carry no arrow: every decision that mattered on the night is either answered by another factor (no plan, no hold point) or contradicted by the recorded data, so an established fatigue would still fail Influence on this record. Whether he was rostered into that position is tested on the roster object.	
Influence:	Not tested
Importance:	Not tested

Individual Actions

Heading eased five degrees to starboard without a decision to turn

Existence:	Pass, virtually certain (100%)
Argument For: From 18:07:06, the second the plotter froze, the heading increased steadily from 195 to 200 degrees over about a minute and was then held. The recorded rate of turn peaks at 6 degrees per minute at 18:07:40, while the master was at the frozen screen (screen taps at 18:07:26, 'frozen again' at 18:07:44), and is back to zero by 18:08:50; rudder angle follows helm demand within 1.5 degrees throughout, so the turn was steered. In interview the master said: 'I eased her to starboard while I was fiddling with the plotter and then looking for four. Five degrees. I never made that decision.' His safety report says the same. Nothing on the bridge showed him the new heading: the plotter's vessel symbol was frozen and the beacon was screened. Argument Against: Nil Reasoning: The recorded heading, rate of turn and rudder angle, and the master's own account, agree.	
Influence:	Pass, virtually certain (100%)
Argument For: This is the action that took the vessel out of the channel. Held at 200 degrees against the channel's 195, it moved the vessel west at about 20 m a minute and across the 20 m to the western edge by about 18:09:00. Argument Against: Nil Reasoning: With no current and no wind effect at that state of tide, the westing that took the vessel out of the dredged channel is this heading change and nothing else.	
Importance:	Not tested

Transit continued at eight knots from the plotter freeze until the shallow alarm

Existence:	Pass, virtually certain (100%)
Argument For: The AIS track records no reduction of speed after the freeze at 18:07:06: 8.4 kt at the freeze and 8.1 kt at 18:08:00, and the fall to 7.1 kt by 18:11:48 is accounted for by shoaling drag with no throttle movement on the engine recorder until 18:11:52. The bridge audio records the master's attention moving to the search for the beacon at 18:08:31, 85 seconds after the freeze, and no consideration of stopping at any point. At 18:10:02 he saw the beacon dead ahead and knew he was off the line to the west. He brought the vessel back to port to regain the channel rather than stopping, because he took himself to be just outside the edge when he was 39 m west of the centreline; the course over ground came back from 200 to 197 degrees, still west of the channel's 195. There was no other traffic in the channel, the water was flat, the wind was offshore at 8 kt and the vessel could have been stopped in 92 m. The master's account, in interview and in his safety report, is that he did not consider stopping when the plotter froze, that he went straight to the visual transit because in his mind the visual transit is the backup, and that after the sighting stopping on the edge of a bank he could not see felt worse than easing back in. No lookout was on the bridge: the deckhand had been sent to collect fares at 18:04:14 and this is the normal practice on the route, which nothing in OPS-3 addresses either way. Argument Against: Nil Reasoning: Recorded track, the engine recorder, the recorded audio and the master's own account agree.	
Influence:	Pass, extremely likely (95%)
Argument For: Continuing at eight knots with no means of fixing the vessel's position is what turned the loss of the plotter into a grounding. Stopping was available from the freeze until about 18:11:40, including the 98 seconds after the master knew he was off the line, and cost nothing. Argument Against: Nil Reasoning: The vessel crossed the channel's western edge 114 seconds after the freeze, and a crash stop begun at any point before about 18:11:40 stops it short of the contact point. A stop in the first 114 seconds leaves it in the channel; a stop after the sighting leaves it on the bank's margin, upright and not holed.	
Importance:	Not tested

Plotter fault passed on verbally at the change of master and not entered in the defect book

Existence:	Pass, virtually certain (100%)
Argument For: The plotter's log records two freezes on the morning shift, at 10:42 and 11:17 local, both of which the unit recovered from by itself. The outgoing master told the relieving master about both at the handover at about 13:05 at the town wharf; the relieving master replied that it had done the same to him the month before. Neither entered it in the wheelhouse defect book, neither reported it to the Operations Manager, and no entry of any kind was made for 07/07/2026 before the occurrence. Both accounts of the handover agree in every particular and both men volunteered the omission before they were asked about it. Argument Against: Nil Reasoning: Established by the defect book, the maintenance system and both masters' own accounts.	
Influence:	Fail, unlikely (15%)
Argument For: An entry at 13:05 would have put the fault in the book a foot from where the relieving master was standing, in about eleven seconds, and nothing required either man to make it. Argument Against: Nil Reasoning: Failed. The freeze at 18:07 was a firmware fault. An entry made at 13:05 would have been reviewed by the Operations Manager at the end of the week, and nothing written that afternoon could have applied firmware to the unit, or taken the vessel off TF18, by 18:07. The defect chain that does reach the freeze runs through the procedure, tested on its own object: a year of freezes that were never written down, of which these two are the last.	
Importance:	Pass, likely (80%)
Argument For: It did not influence this occurrence and it is still a finding, because it is the practice and not a lapse: both masters made the same omission, both volunteered it, and both described the same rule for it. A cleared fault on the vessel's primary means of position fixing passed on by word of mouth leaves the next master knowing and everyone ashore not, and the next fault passed on that way may be one that an entry on the day would catch. Argument Against: Nil Reasoning: The instance of the procedure gap, worth addressing with it rather than as a matter for the two men.	

Engines stopped and no attempt made to refloat on a falling tide

Existence:	Pass, virtually certain (100%)
Argument For: Both engines were stopped 27 seconds after contact and were not restarted. The master's pan pan at 18:13:04 states 'I am not attempting to refloat' as a decision already made, 60 seconds after the vessel came to rest. He directed the No. 2 starboard void to be sounded, reported 75 mm of ingress to Port Control at 18:17:05 and had it sounded ten minutely thereafter. Passengers were addressed within 36 seconds of the impact and again at 18:14:52, and lifejackets were issued to all 38. The survey later confirmed a 180 mm split in the starboard hull plating in way of that void, with ingress reaching approximately 260 mm by the time the vessel refloated on the tide at 00:20. His reasoning in interview: falling tide, unknown bottom, thirty eight people aboard, and a vessel that cannot sink while it is on the bank. Argument Against: Nil Reasoning: Recorded audio and the engine recorder throughout, with the survey confirming the breach.	
Influence:	Pass, very likely (90%)
Argument For: The vessel was holed. Backing a holed vessel off a bank into the channel on a falling tide, at night, with 38 passengers aboard and no assessment of the damage, is the decision that turns a grounding into something else. That decision was never taken, and the decision against it was made inside a minute. Argument Against: Nil Reasoning: Recorded as a decrease in risk. The breach was in the compartment the vessel was sitting on; refloating it into 3.2 m of water with the ingress unassessed would have put a flooding compartment into deep water with the passengers still aboard.	
Importance:	Not tested

Occurrence Events

Steering or propulsion failure before the vessel left the centreline

Existence:	Fail, very unlikely (5%)
Argument For: Hypothesis tested because a five degree heading change followed by a steady set out of the channel is what a jammed rudder or a lost engine looks like from outside. It fails, on two independent grounds, and what refutes it is measurement rather than argument. The engine recorder gives both engines at 1,780 and 1,784 rpm within 12 rpm of each other from 18:03 to 18:11:52 with no alarm, no excursion and no throttle movement; steering gear hydraulic pressure holds continuously at 96 to 99 bar against a normal band of 90 to 105; and rudder angle follows helm demand within 1.5 degrees throughout. The rate of turn on the recorder peaks at 6 degrees per minute and returns to zero, which is a heading applied and then held, not a sheer. Separately, the whole steering system was removed and function tested on 12/07/2026 and every component was serviceable with no internal leakage and no sign of a pre-existing defect. The bridge audio contains no steering or engine remark at any point. Argument Against: Nil Reasoning: The recorder traces, the shape of the turn and the post-occurrence examination agree. The starboard shaft, propeller and rudder damage all date from the contact, not before it. The 5% is the standing of the hypothesis and not of the traces, which are themselves settled.	
Influence:	Not tested
Importance:	Not tested

Vessel left the dredged channel on the bank side

Existence:	Pass, virtually certain (100%)
Argument For: The vessel was 1 m west of the charted centreline abeam the No. 2 beacon at 18:05:30 and held it until 18:07:06. From that time the heading increased steadily from 195 to 200 degrees over about a minute and was then held, and the vessel set west at about 20 m per minute. It reached the channel's western edge at about 18:09:00, 0.40 nm north of the No. 4 beacon, was 39 m west of the centreline by 18:10:02 and over Boolarong Bank from about 18:11:00. The recorded rate of turn peaks at 6 degrees per minute at 18:07:40 and is back to zero by 18:08:50, which is a heading change applied and held rather than a current or wind effect; the wind was offshore at 8 kt and there is no cross-set in the channel at that state of tide. Argument Against: Nil Reasoning: Recorded track, corroborated by the surveyed grounding position and the physical marks on the bed.	
Influence:	Pass, virtually certain (100%)
Argument For: Nothing else on this map matters without this. The channel is 40 m wide and dredged to 3.2 m, and the bank on its western side dries in places; once outside the western edge the vessel was over water that shoals at about 1 in 70. Argument Against: Nil Reasoning: The grounding follows directly and only from the vessel being outside the dredged channel.	
Importance:	Not tested

Shallow alarm at its 2019 setting of 0.9 m gave sixteen seconds' warning

Existence:	Pass, virtually certain (100%)
Argument For: The echo sounder's shallow alarm is set to 0.9 m under the keel, the value written into the setup at commissioning in 2019 and never changed. It sounded once on the voyage, at 18:11:48, and the bridge audio records the tone beginning at that second. Both throttles were moved to astern together 4 seconds later and contact followed at 18:12:04, 16 seconds after the alarm. At 7.1 kt the vessel covered 58 m in that time; at the surveyed bank gradient of about 1 in 70 across that section the alarm could not have sounded appreciably earlier at that setting, and the vessel's own trials figure for a crash stop from 8 kt is 92 m. Astern thrust was developing but had not reached full at the moment of contact. Argument Against: Nil Reasoning: The alarm log, the bridge audio and the recorded speed all agree to the second.	
Influence:	Pass, extremely likely (95%)
Argument For: The last defence the vessel had gave sixteen seconds against a stopping distance it could not achieve in them. That is a finding about the setting and the speed, not about the equipment, which did exactly what it was set to do. Argument Against: Nil Reasoning: Sixteen seconds at seven knots is 58 m against a 92 m crash stop. On the surveyed 1 in 70 slope a setting of about 1.6 m under the keel would have sounded about 107 m short of the contact, enough for a crash stop with four seconds to react, and still would not sound in the dredged channel at low water. A setting that gave forty seconds would, which is why the setting has to be chosen for the channel rather than simply raised.	
Importance:	Not tested

Grounding on Boolarong Bank with 38 passengers embarked

Existence:	Pass, virtually certain (100%)
Argument For: MV Tarrakin Star, a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers, made contact with Boolarong Bank at 6.8 kt over the ground at 18:12:04 local on 07/07/2026 while inbound on service TF18, and came to rest within 11 m, heeled 4 degrees to starboard, 74 m west of the charted centreline of the Merrandi Channel and 54 m outside the dredged channel, on a falling tide in darkness. 38 passengers, a master, a deckhand and a kiosk attendant were on board. Furrows consistent with the starboard hull and skeg run 11 m from the point of first contact on a heading of 197 degrees. Argument Against: Nil Reasoning: Nil	
Influence:	Not tested
Argument For: Substantial damage: starboard hull bottom plating set up over 4.1 m with a 180 mm split in way of the No. 2 void and progressive flooding of that void to approximately 260 mm by the time the vessel refloated; starboard propeller blades bent and the shaft bent 1.4 mm; starboard rudder stock distorted 3 degrees; starboard skeg torn away. Nil serious injuries; one passenger reported a grazed shin and two were treated at the wharf for distress. All 38 passengers were transferred ashore by the port workboat by 19:26. The vessel refloated on the tide at 00:20 and was withdrawn from service. The Merrandi Channel was closed to commercial traffic for 19 hours. The vessel came to rest where the bed is 0.50 m below chart datum; 30 m further west the bed is 0.35 m higher again, and there it would have taken the ground harder and taken a larger list. Argument Against: Nil Reasoning: Nil	
Importance:	Not tested